

CHAPTER 6

Noah Parker

6.1. (a) Let M be a Noetherian A -module and $u : M \rightarrow M$ a module homomorphism.

If u is surjective, then it is an isomorphism.

(b) If M is Artinian and u is injective, then again u is an isomorphism.

Proof. (a) Consider the filtration $\ker u \subset \ker u^2 \subset \cdots$, and the associated complex of modules

$$\cdots \rightarrow \frac{\ker u^2}{\ker u} \xrightarrow{u} \ker u \rightarrow 0.$$

Note that $u : M \rightarrow M$ is surjective $u^{-1}(\ker u^k) = \ker u^{k+1}$, so each of the above maps is surjective. It follows by induction that the complex is zero if and only if it is bounded. M Noetherian gives us $n > 0$ such that $\ker u^n = \ker u^{n+1}$, whence the complex is bounded, and so $\ker u = 0$ as desired.

(b) Consider the filtration $\cdots \subset \operatorname{im} u^2 \subset \operatorname{im} u$, and the associated complex of modules

$$0 \rightarrow \operatorname{cok} u \xrightarrow{u} \frac{\operatorname{im} u}{\operatorname{im} u^2} \rightarrow \cdots$$

The kernel of each map is of the form $u^{-1}(\operatorname{im} u^{k+1}) = u^{-1}(u(\operatorname{im} u^k)) = \operatorname{im} u^k$ by injectivity of u , so each map is injective. It follows by induction that the complex is zero if and only if it is bounded. M Artinian gives us $n > 0$ such that $\operatorname{im} u^n = \operatorname{im} u^{n+1}$, whence the complex is bounded, and so $\operatorname{cok} u = 0$ as desired. $\square$

6.2 Let M be an A -module. If every non-empty set of finitely generated submodules of M has a maximal element, then M is Noetherian.

Proof. Let Σ be a non-empty set of submodules of M , and let Σ' be the set of finitely generated submodules of elements of Σ . $\square$

6.4. Let M be a Noetherian A -module, and let $\mathfrak{a}$ be the annihilator of M in A . Prove that $A/\mathfrak{a}$ is a Noetherian ring.

If we replace *Noetherian* by *Artinian*, is this result still true?

Proof. $\square$

6.11. Let $f : A \rightarrow B$ be a ring homomorphism and suppose that $\operatorname{Spec} B$ is a noetherian space. Prove that $f^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is a closed mapping if and only if f has the going up property (5, Ex. 10).

Proof. $\square$

6.12 Let A be a ring such that $\operatorname{Spec} A$ is Noetherian. Show the set of prime ideals of A satisfies the ascending chain condition. Is the converse true?

Proof. $\square$